FinTech & Digital Economy

By

Group B

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DECLARATION

We certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of our knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in the text.

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Content

<u>FinTech & Digital Economy</u>

FinTech refers to the intersection of finance and technology, using digital platforms and tools to provide financial services.

Examples:

• Mobile Payments – Apple Pay, Google Pay, M-Pesa

• Digital Wallets – PayPal, Venmo, Cash App

• Peer-to-Peer Lending – Lending Club, Funding Circle

• Robo-Advisors – Betterment, Wealth front

• Cryptocurrencies – Bitcoin, Ethereum (Blockchain technology)

Digital Economy

The Digital Economy refers to economic activity driven by digital technologies such as the internet, mobile devices, and social media.

Examples:

• E-commerce – Amazon, eBay, Shopify

• Digital Marketplaces – Uber, Airbnb, Freelancer

• Online Banking – Manage finances digitally

• Digital Content – Netflix, Spotify, Apple Music

• Gig Economy – TaskRabbit, Fiverr, Upwork

Key Benefits of FinTech & Digital Economy

- Increased Accessibility – Greater access to financial and economic opportunities
- Improved Efficiency – Automation reduces cost and time
- Enhanced Convenience – Access services anytime, anywhere
- New Business Models – Peer-to-peer lending, gig economy platforms

Impact on Society and Economy

- Financial Inclusion – Broader access for underserved populations
- Economic Growth – Encourages innovation and new ventures
- Job Creation – Expands opportunities in tech and gig sectors

<u>Blockchain beyond crypto</u>

Blockchain is not limited to Bitcoin or other cryptocurrencies. Its real potential lies in solving **trust, transparency, and efficiency problems** across industries. The technology is a **distributed, immutable, and transparent ledger** where multiple parties can share and verify data **without central control**.

Below we explore **three major real-world applications**:

- ❑ Supply Chain Management
- ❑ Health Records
- ❑ Land Registries

Supply Chain Management

Current Challenges

- Complex networks with many intermediaries.
- Counterfeit goods and fraud.
- Delays due to paperwork and manual tracking.
- Customers unable to verify authenticity or ethical sourcing.

Blockchain Solution

- Creates a **single, shared, tamper-proof record** of every step (farm → factory → warehouse → retail).
- Provides **end-to-end transparency** for all stakeholders.
- Smart contracts automatically **trigger payments/deliveries** when conditions are met.
- Real-time tracking improves efficiency and safety.

Example

- **IBM Food Trust (Walmart)**: Track food products from farm to shelf; contamination sources can be identified in seconds instead of weeks.

Health Records

Current Challenges

- Records are stored in **silos** across different hospitals and clinics.
- High risk of **tampering, hacking, or data loss**.
- Patients have little control over their own data.
- Difficult to share medical history across regions or countries.

Blockchain Solution

- Patient-centered records stored securely on blockchain.
- Patients control who can access their health data and can grant/revoke access anytime.
- Records are encrypted, immutable, and timestamped.
- Reduces duplicate testing, fraud in insurance claims, and administrative inefficiencies.

Example

- Estonia e-Health: All citizens' health records are blockchain-secured, and patients can track who accessed their data and why.

Land Registries

Current Challenges

- Land ownership documents often stored in paper form.
- High risk of corruption, manipulation, and forgery.
- Disputes over ownership take years to resolve.
- Transactions involve expensive legal processes

Blockchain Solution

- Land titles stored on blockchain are tamper-proof and transparent.
- Transfers of ownership are simplified using smart contracts.
- Reduces fraud, speeds up transactions, and ensures citizens’ rights.

Example

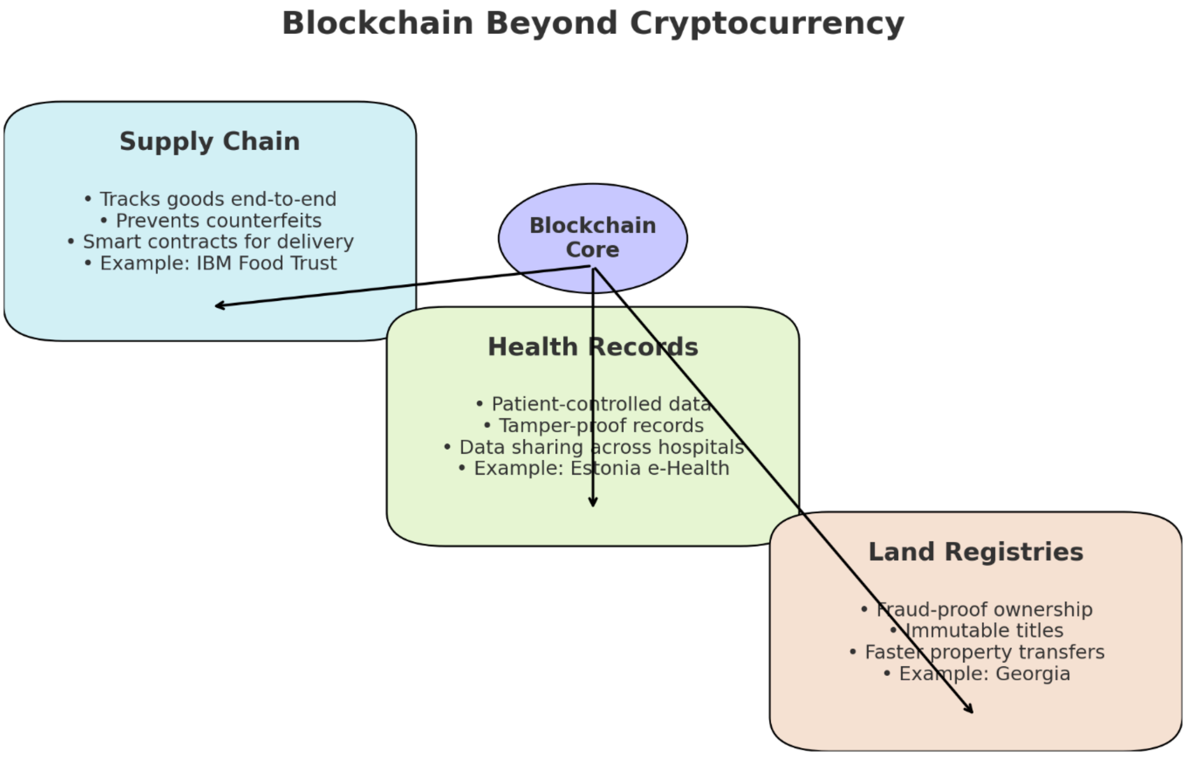
- Georgia Land Registry (since 2016): Uses blockchain to secure property records, reducing corruption and improving trust.
- Sweden and India are also piloting similar systems.

Advantages of Blockchain Beyond Cryptocurrency

- Trust without Intermediaries: Reduces dependency on central authorities.
- Enhanced Transparency and Accountability: Every transaction is visible to authorized participants.
- Fraud Reduction: Immutable records prevent unauthorized alterations.
- Efficiency and Cost Savings: Automates processes, reduces paperwork, and lowers operational costs.
- Global Accessibility: Can enable secure financial and administrative systems in emerging markets.

Challenges and Considerations

- Scalability: High-volume transaction networks may face delays.
- Regulation: Legal frameworks for blockchain-based systems are still evolving.
- Integration: Legacy systems may require significant upgrades to integrate blockchain.
- Energy Consumption: Some blockchain networks, especially proof-of-work-based, consume substantial energy.



Blockchain Beyond Cryptocurrency
Supply Chain
• Tracks goods end-to-end
• Prevents counterfeits
• Smart contracts for delivery
• Example: IBM Food Trust
Blockchain Core
Health Records
• Patient-controlled data
• Tamper-proof records
• Data sharing across hospitals
• Example: Estonia e-Health
Land Registries
• Fraud-proof ownership
• Immutable titles
• Faster property transfers
• Example: Georgia

Central Bank Digital Currencies (CBDCs) in the Evolution of FinTech

The global financial system is undergoing a profound transformation driven by digital innovation. The emergence of cryptocurrencies and private stablecoins has challenged traditional monetary systems, prompting central banks worldwide to explore the development of their own digital currencies. A Central Bank Digital Currency (CBDC) is a digital form of sovereign currency that represents a direct liability of the central bank, potentially combining the digital convenience of private payment systems with the safety and stability of central bank money (BIS, 2023). For a developing economy like Sri Lanka, which is navigating a path toward economic stabilization and modernization, understanding the CBDC phenomenon is not merely academic but strategically critical. The potential to bank the unbanked, reduce transaction costs, and create a more resilient financial infrastructure presents a compelling opportunity. However, the risks of technological failure, privacy erosion, and disruption to the commercial banking sector are substantial and require careful mitigation.

Research Questions and Document Structure

This systematic review aims to synthesize existing literature to address the following research questions:

1 Global Trends and Models What are the dominant global trends and architectural models in CBDC development?

2 Benefits and Risks What are the empirically supported benefits and risks associated with CBDC implementation?

3 Literature Gaps What specific gaps exist in the literature regarding CBDC deployment in developing economies like Sri Lanka?

4 Strategic Pathway What evidence-based strategic pathway can be recommended for Sri Lanka?

The paper is structured as follows: Section 2 outlines the methodology, Section 3 presents a thematic synthesis of the literature, Section 4 provides a critical discussion, Section 5 identifies research gaps and future directions, and Section 6 concludes

Systematic Review Approach

Data Sources and Search Strategy Electronic searches were performed in Scopus, IEEE Xplore, Google Scholar, and the Bank for International Settlements (BIS) repository. The search covered the period from 2015 to 2025 to capture both foundational concepts and recent developments.

Keywords and Search Strings

("Central Bank Digital Currency" OR CBDC) AND (design OR framework OR implementation) ("digital rupee" OR "digital currency") AND ("financial inclusion" OR "monetary policy") (CBDC) AND (risk OR challenge) AND (privacy OR "bank disintermediation")

Inclusion and Exclusion Criteria Inclusion Criteria

Peer-reviewed journal articles, conference proceedings, and official reports from central banks and international financial institutions (IMF, BIS). The focus was on literature published in English, with a preference for studies from 2018 onwards. Exclusion Criteria: Non-peer-reviewed blogs, news articles, and opinion pieces were excluded unless they provided unique data or official announcements from a central bank.

Article Selection

The initial search yielded over 200 documents. After removing duplicates and screening titles and abstracts for relevance, 75 full-text articles were assessed for eligibility. A final set of 38 sources was selected for in-depth analysis based on their direct relevance to the research questions and methodological rigor. The analysis followed a thematic synthesis approach, coding the literature into emergent themes.

Global Landscape and Design Models

The analyzed literature was synthesized into four central themes. Research shows that CBDC development has progressed from theoretical concept to active experimentation. Brunner Meier & Landau (2022) contextualize this as part of the "digital clash" between public and private money.

Pioneering Retail CBDCs - The Bahamas' 'Sand Dollar' (2020) is widely cited as the first fully deployed retail CBDC, designed to address financial inclusion across its archipelago (Central Bank of The Bahamas, 2020).

Adoption Challenges - Nigeria's 'e-Naira' launch in 2021 provided critical lessons on the challenges of user adoption, highlighting that technology alone is insufficient without public trust and clear use cases (Oyedele, 2022).

Scalability and Integration - China's large-scale pilot of the Digital Currency Electronic Payment (DCEP) system, involving hundreds of millions of users, demonstrates the scalability and integration of CBDCs within a complex digital economy (Auer et al., 2022).

Benefits and Risks

Benefits and Opportunities

- Financial Inclusion - Digital currencies can significantly reduce barriers to financial services for rural and low-income populations (Agarwal & Zhang, 2021).
- Payment System Efficiency - CBDCs can drastically reduce the cost and time of remittances, a key source of foreign exchange for Sri Lanka (BIS Innovation Hub, 2023).
- Fiscal Transparency - They could enable targeted and transparent distribution of social benefits, reducing graft and leakage (Kiff et al., 2020).

Risks and Challenges

- Privacy and Surveillance - The potential for state-led financial surveillance is the most frequently cited risk. Without strong legal safeguards, CBDCs could fundamentally erode financial privacy (Human Rights Foundation, 2024).
- Financial Disintermediation - CBDCs could trigger digital bank runs, where depositors flee commercial banks for the safety of the central bank during a crisis (Kumho & Noone, 2018).

Strategic Implications for Sri Lanka

This review reveals a clear dichotomy between the rapid advancement of CBDCs in theory and the cautious, problem-plagued reality of their implementation. While the theoretical benefits for a country like Sri Lanka are substantial4particularly in leapfrogging legacy payment infrastructure4the practical risks are non-trivial.

The literature strongly suggests that a successful CBDC is not defined by its technological novelty but by its ability to solve specific national problems. For Sri Lanka, the primary objective should be financial inclusion and payment efficiency, not monetary policy experimentation.

The discussion also highlights a critical weakness in the global literature: a predominant focus on large or advanced economies, with a scarcity of tailored analyses for lower-middle-income countries facing unique challenges in infrastructure, governance, and public trust.

The hybrid architectural model emerges as the most pragmatically recommended design, offering Sri Lanka a path to avoid the scalability limits of pure DLT while maintaining sufficient decentralization to build trust. However, the success of any model is contingent on establishing an independent data privacy regime, a point critically underemphasized in many technical proposals. The experiences of Nigeria and the Bahamas underscore that public trust is the ultimate currency for a CBDC's success, a factor that must be cultivated through transparency and robust public education campaigns in the Sri Lankan context.

Research Gaps and Future Directions

This synthesis identifies several critical gaps in the existing research, pointing to clear avenues for future inquiry

Context-Specific Implementation Frameworks - Future research should focus on creating adaptable models for nations like Sri Lanka, addressing challenges such as intermittent connectivity and lower digital literacy.

Cross-Border Payment Integration - Little research exists on how smaller economies like Sri Lanka could integrate with regional payment systems (e.g., with India or ASEAN nations) to reduce remittance costs.

Public Perception and Behavioral Economics - Research is needed to survey Sri Lankan citizens' concerns and preferences regarding digital currency features, privacy, and accessibility.

Cybersecurity and Resilience Models - Future work should develop and test cost-effective cybersecurity and operational resilience frameworks specifically designed for the resource constraints of a central bank like the Central Bank of Sri Lanka (CBSL).

Conclusion and Strategic Pathway

This systematic review has synthesized the current body of knowledge on CBDCs, demonstrating their potential to act as a catalyst for financial modernization while also underscoring their significant risks. The global trend is unequivocal: digital currency is becoming a central pillar of the future monetary system. For Sri Lanka, the path forward must be one of strategic, evidence-based progression, not haste. The findings of this review suggest that the country should prioritize the development of a retail CBDC with a hybrid architecture, placing privacy protection and financial stability safeguards at the core of its design. The primary contribution of this review is to consolidate a dispersed field of research into a coherent framework and to highlight the specific research and policy prerequisites for Sri Lanka. By proceeding with caution, learning from global pioneers, and investing in foundational research, Sri Lanka can position itself to harness the benefits of the digital currency revolution while mitigating its inherent dangers. The success of a future Sri Lankan Digital Rupee will ultimately depend not on its technical sophistication, but on its ability to earn and maintain the trust of the people it is designed to serve.

Artificial Intelligence (AI) in Microfinance and Credit Scoring

AI leverages alternative data sources to assess creditworthiness, especially in regions where formal banking data is scarce.

Microfinance Applications

- AI algorithms analyze mobile phone usage, social networks, and transaction patterns to approve loans for low-income individuals.
- Reduces dependency on traditional collateral-based lending.

Rural Credit Scoring

- Farmers and micro-entrepreneurs receive AI-driven risk assessments, enabling access to loans previously unavailable to them.
- Integrated mobile apps provide automated reminders, loan recommendations, and personalized financial literacy programs.

Example: Tala uses smartphone data to provide microloans in developing countries, targeting populations with limited access to traditional banking.

Benefits

- Expands access to finance.
- Reduces lender risk via data-driven scoring.
- Promotes financial literacy and responsible borrowing.

APIs Enabling New Financial Ecosystems

In the digital era, Application Programming Interfaces (APIs) have become a driving force behind innovation in the financial industry.

APIs act as secure bridges that allow different systems—such as banks, fintech companies, and third-party applications—to communicate and share data efficiently.

This connectivity has created new financial ecosystems, transforming traditional banking into open, interconnected networks.

What Are APIs?

An API (Application Programming Interface) is a software intermediary that enables two applications to communicate with each other.

In the context of finance, APIs allow financial institutions to share customer data, with consent, to develop new services and improve customer experience.

Example:

A mobile app can use a bank’s API to show a customer’s account balance, process payments, or even recommend budgeting plans.

Role of APIs in Financial Ecosystems

APIs are at the heart of modern Open Banking and Fintech innovation.

They enable:

- Data sharing: Banks can securely share data with authorized third- party apps.
- Integration: Fintechs and startups can connect with banks’ systems easily.
- Automation: APIs simplify transactions and reduce manual processing.
- Innovation: Developers can build personalized and customer-focused financial solutions.

Benefits of APIs in Finance

- Enhanced Customer Experience: APIs provide faster, more personalized digital services.
- Transparency: Customers gain more control over their financial data.
- Cost Efficiency: Automation and integration reduce operational costs.

- Innovation & Competition: APIs encourage fintech to develop new solutions that challenge traditional banking.

Example :

1. Online Banking App

When you use a banking app on your phone to check your balance or send money — that app connects to your bank’s system using an API.

→ You get real-time information quickly and securely.

2. Fintech Apps (like loan or payment apps)

Some apps, like digital wallets or loan apps, use bank APIs to access your financial data (with permission).

→ This allows you to pay bills, transfer money, or even apply for loans without visiting the bank.

3. Embedded Finance

When you buy something online and choose to pay in installments (“Buy Now, Pay Later”), an API connects the e-commerce site with a finance company.

→ The process happens instantly inside the app or website.

Challenges

Despite the advantages, API adoption also presents some challenges:

- Data Security Risks: Protecting sensitive financial data is critical.
- Regulatory Compliance: Banks must follow strict data protection regulations.
- Integration Complexity: Older banking systems may not easily connect with new APIs.
- Customer Trust: Building confidence in data-sharing systems remains essential.

Future of APIs in Finance

Over the next five years, APIs will continue to shape the financial world through:

- AI-driven personalized finance (custom recommendations, insights).
- Banking-as-a-Service (BaaS): Allowing non-banking firms to offer financial services.
- Cross-border APIs: Enabling global and instant payments.
- Open Finance Expansion: Connecting banking with insurance, investment, and telecom services.

APIs will make banking more open, customer-centric, and technology-driven.

Conclusion

APIs are transforming the financial industry by enabling a connected, innovative, and inclusive ecosystem.

They empower banks and fintech companies to collaborate, create new digital services, and offer customers more convenience and control over their finances.

In the future, APIs will remain the foundation of Open Finance, leading the way toward a smarter and more integrated financial world.

Conclusion

The **FinTech revolution** has fundamentally transformed the global financial landscape, enabling faster, more transparent, and inclusive financial services. By integrating advanced technologies such as **blockchain, artificial intelligence (AI), Central Bank Digital Currencies (CBDCs), and Open Banking APIs**, the financial sector is evolving from traditional, siloed systems into **connected, efficient, and customer-centric ecosystems**.

Key Takeaways

1. **Blockchain Beyond Cryptocurrency**

   Blockchain has expanded its applications beyond digital currencies to areas such as supply chain management, healthcare records, and land registries. Its decentralized and immutable nature ensures transparency, security, and traceability, enabling organizations and governments to build trust and efficiency into their operations.

2. **Central Bank Digital Currencies (CBDCs)**

   CBDCs represent the evolution of national currencies into fully digital forms, combining the benefits of FinTech innovations with government oversight. They enhance financial inclusion, reduce transaction costs, enable real-time settlements, and integrate seamlessly with mobile wallets and other digital platforms.

3. **AI in Microfinance and Credit Scoring**

   AI-powered financial tools are democratizing access to credit and banking services, particularly in rural and underserved areas. By analyzing alternative data sources, AI enables personalized lending, accurate credit scoring, and financial literacy, empowering individuals and small businesses alike.

4. **Open Banking APIs**

   Open Banking allows secure and consent-driven sharing of financial data between banks, fintech apps, and third-party providers. This fosters innovation, promotes competition, and creates new integrated financial ecosystems where consumers can manage payments, budgeting, lending, and investments from a single platform.

Challenges and Considerations

While FinTech promises numerous benefits, challenges remain. Regulatory compliance, cybersecurity, digital literacy, data privacy, and infrastructure costs must be carefully managed to ensure secure, reliable, and equitable access to digital financial services.

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